



FRAUD DETECTION AND TRANSACTION ANALYSIS USING PYTHON



Python



Pandas

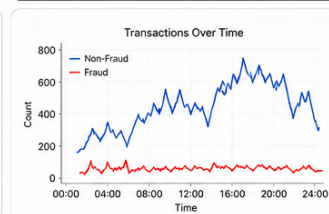
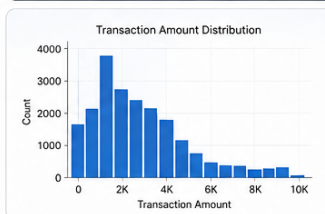
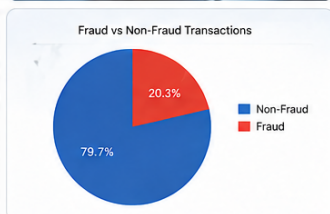


Matplotlib



Project Overview

This project analyzes financial transaction data to identify patterns and factors associated with fraudulent activities using Python, Pandas for data analysis and Matplotlib for data visualization.



Goals and Outcomes

The main objective of this project is to analyze financial transaction data and identify patterns that may indicate fraudulent activity. The project aims to clean and preprocess the dataset, handle missing values and inconsistencies, perform descriptive analysis, compare fraudulent and legitimate transactions, and visualize important transaction and risk-related factors using Pandas and Matplotlib. The analysis will help understand which characteristics are commonly associated with fraudulent transactions and provide meaningful insights from the dataset.

Project Scope

Financial Fraudulent Transaction Analysis Using Python is a data analysis project focused on identifying and understanding patterns associated with fraudulent financial transactions. The project uses Python and Pandas to clean, organize, and analyze transaction data, while Matplotlib is used to create visualizations that help identify relationships between factors such as transaction amount, transaction time, login attempts, merchant risk score, IP risk score, transaction distance, and fraudulent activity. The analysis aims to provide insights into the characteristics and patterns of fraudulent transactions.

1 Dataset Loading and Initial Overview

1.1 Dataset Loading

```
import pandas as pd

from google.colab import files
uploaded = files.upload()

Saving Fraud_Detection_Dataset.csv to Fraud_Detection_Dataset.csv

df=pd.read_csv('Fraud_Detection_Dataset.csv')

df.head()
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
0	TXN1000000	CUST46048	Axis Bank	Current	Bharat Petroleum	Pharmacy	3998.30
1	TXN1000001	CUST65392	Canara Bank	Salary	Bharat Petroleum	Restaurant	2706.99
2	TXN1000002	CUST19116	State Bank of India	Current	Swiggy	Pharmacy	3196.37
3	TXN1000003	CUST70589	Union Bank of India	Salary	BookMyShow	Restaurant	2090.97
4	TXN1000004	CUST28301	Axis Bank	Salary	Jio	Telecom	2747.35

Output columns 2 of 4

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
0	INR	IMPS	Recharge	...	2	103	4313.87
1	INR	Credit Card	Purchase	...	2	106	2948.79
2	INR	RTGS	Purchase	...	0	77	2084.17
3	INR	Credit Card	Purchase	...	3	50	2518.13
4	INR	Net Banking	Bill Payment	...	3	76	2244.80

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
0	125217.62	3561.0	10.52	60.11	1770.18	5.23
1	18938.41	2575.0	54.85	4.67	2434.39	156.70
2	772522.81	2601.0	81.80	9.77	1710.58	56.14

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
3	166749.36	3624.0	58.93	84.23	1124.39	31.11
4	285724.28	1918.0	31.69	14.09	2005.49	91.58

Output columns 4 of 4

	fraud_label
0	Fraudulent
1	Legitimate
2	Legitimate
3	Legitimate
4	Legitimate

df.tail()

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
12802	TXN1012802	CUST83165	Punjab National Bank	Salary	DMart	Restaurant	1504.86
12803	TXN1012803	CUST92904	Bank of Baroda	Business	Airtel	Grocery	7106.29
12804	TXN1012804	CUST43977	ICICI Bank	Salary	Indian Oil	Shopping Mall	4187.23
12805	TXN1012805	CUST37643	Union Bank of India	Salary	Reliance Fresh	Fuel Station	3207.46
12806	TXN1012806	CUST12219	Canara Bank	Savings	Swiggy	Telecom	2868.06

Output columns 2 of 4

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
12802	INR	IMPS	Recharge	...	3	28	2028.88
12803	INR	Debit Card	Bill Payment	...	2	14	9333.58
12804	INR	Mobile Wallet	Purchase	...	3	62	4700.02
12805	INR	Credit Card	Recharge	...	3	100	3308.09
12806	INR	Mobile Wallet	Ticket Booking	...	3	16	3135.45

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
12802	119589.43	3346.0	34.76	12.99	487.80	7.01
12803	77331.13	1088.0	40.24	76.24	1193.40	155.80
12804	108584.33	2309.0	74.73	74.36	1765.70	54.71
12805	504981.03	2063.0	31.68	29.43	1122.40	24.09
12806	618620.82	2333.0	46.84	99.86	2242.15	183.95

Output columns 4 of 4

	fraud_label
12802	Legitimate
12803	Legitimate
12804	Legitimate
12805	Fraudulent
12806	Fraudulent

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12807 entries, 0 to 12806
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   transaction_id                        12807 non-null  object
1   customer_id                          12807 non-null  object
2   bank_name                           12807 non-null  object
3   account_type                        12807 non-null  object
4   merchant_name                       12807 non-null  object
5   merchant_category                   12807 non-null  object
6   transaction_amount                  12807 non-null  float64
7   currency                           12807 non-null  object
8   payment_method                     12807 non-null  object
9   transaction_type                    12807 non-null  object
10  transaction_hour                    12807 non-null  int64
11  login_attempts                     12807 non-null  int64
12  failed_login_count                 12807 non-null  int64
13  previous_transactions              12807 non-null  int64
14  average_transaction_amount         12807 non-null  float64
15  account_balance                    12807 non-null  float64
16  beneficiary_age_days              12803 non-null  float64
17  merchant_risk_score                12807 non-null  float64
18  ip_risk_score                      12807 non-null  float64
19  transaction_distance_km            12807 non-null  float64
20  transaction_duration_sec           12807 non-null  float64
21  fraud_label                        12807 non-null  object
dtypes: float64(8), int64(4), object(10)
memory usage: 2.1+ MB
```

```
df.describe(include='all')
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
count	12807	12807	12807	12807	12807	12807	12807.000000
unique	12807	11953	10	4	18	11	NaN
top	TXN1012806	CUST36978	State Bank of India	Current	DMart	Fuel Station	NaN
freq	1	3	1344	3224	750	1219	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	3363.525143
std	NaN	NaN	NaN	NaN	NaN	NaN	2006.498548
min	NaN	NaN	NaN	NaN	NaN	NaN	70.870000
25%	NaN	NaN	NaN	NaN	NaN	NaN	1881.435000
50%	NaN	NaN	NaN	NaN	NaN	NaN	2963.690000
75%	NaN	NaN	NaN	NaN	NaN	NaN	4419.395000
max	NaN	NaN	NaN	NaN	NaN	NaN	15646.070000

Output columns 2 of 4

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
count	12807	12807	12807	...	12807.000000	12807.000000	12807.000000
unique	1	8	6	...	NaN	NaN	NaN
top	INR	NEFT	Ticket Booking	...	NaN	NaN	NaN
freq	12807	1677	2223	...	NaN	NaN	NaN
mean	NaN	NaN	NaN	...	1.492621	60.295932	3352.838403
std	NaN	NaN	NaN	...	1.122410	34.804537	2189.521192
min	NaN	NaN	NaN	...	0.000000	0.000000	54.800000
25%	NaN	NaN	NaN	...	0.000000	31.000000	1768.290000
50%	NaN	NaN	NaN	...	1.000000	60.000000	2858.220000

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
75%	NaN	NaN	NaN	...	2.000000	90.000000	4359.805000
max	NaN	NaN	NaN	...	3.000000	120.000000	18091.760000

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
count	12807.000000	12803.000000	12807.000000	12807.000000	12807.000000	12807.000000
unique	NaN	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN	NaN	NaN
mean	403026.541509	1993.080294	52.918592	50.083708	1246.636607	152.244809
std	230369.604871	1151.618887	27.432626	28.718411	723.743727	84.961508
min	505.250000	2.000000	5.000000	0.000000	0.040000	5.030000
25%	205161.115000	1005.500000	29.420000	25.365000	611.790000	78.545000
50%	406067.920000	1973.000000	53.010000	49.690000	1244.540000	151.620000
75%	600233.290000	2985.000000	76.690000	74.790000	1880.710000	225.585000
max	799962.120000	3999.000000	100.000000	99.980000	2499.650000	299.980000

Output columns 4 of 4

	fraud_label
count	12807
unique	2
top	Legitimate
freq	11800
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

2 Data Preprocessing

2.1 Handling Missing Values

#Identify all missing values from the dataset
`df.isnull().sum()`

	0
transaction_id	0
customer_id	0
bank_name	0
account_type	0
merchant_name	0
merchant_category	0

	0
transaction_amount	0
currency	0
payment_method	0
transaction_type	0
transaction_hour	0
login_attempts	0
failed_login_count	0
previous_transactions	0
average_transaction_amount	0
account_balance	0
beneficiary_age_days	4
merchant_risk_score	0
ip_risk_score	0
transaction_distance_km	0
transaction_duration_sec	0
fraud_label	0

2.2 To Remove Null Values From the Dataset

```
df['beneficiary_age_days'] = df['beneficiary_age_days'].fillna(0)
```

2.3 Identify all Missing Values From the Dataset

```
df.isnull().sum()
```

	0
transaction_id	0
customer_id	0
bank_name	0
account_type	0
merchant_name	0
merchant_category	0
transaction_amount	0
currency	0
payment_method	0
transaction_type	0
transaction_hour	0
login_attempts	0
failed_login_count	0
previous_transactions	0
average_transaction_amount	0
account_balance	0
beneficiary_age_days	0
merchant_risk_score	0
ip_risk_score	0

	0
transaction_distance_km	0
transaction_duration_sec	0
fraud_label	0

2.4 To Check Whether the Dataset Contains Any More Missing Value After Cleaning

```
df.isnull().values.sum()
```

```
np.int64(0)
```

2.5 To Check the Dimensions of a DataFrame

```
df.shape
```

```
(12807, 22)
```

2.6 To Display the Column Names

```
df.columns
```

```
Index(['transaction_id', 'customer_id', 'bank_name', 'account_type',
      'merchant_name', 'merchant_category', 'transaction_amount', 'currency',
      'payment_method', 'transaction_type', 'transaction_hour',
      'login_attempts', 'failed_login_count', 'previous_transactions',
      'average_transaction_amount', 'account_balance', 'beneficiary_age_days',
      'merchant_risk_score', 'ip_risk_score', 'transaction_distance_km',
      'transaction_duration_sec', 'fraud_label'],
      dtype='object')
```

2.7 To Delete A Column

```
df.drop(columns=['account_balance'])
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
0	TXN1000000	CUST46048	Axis Bank	Current	Bharat Petroleum	Pharmacy	3998.30
1	TXN1000001	CUST65392	Canara Bank	Salary	Bharat Petroleum	Restaurant	2706.99
2	TXN1000002	CUST19116	State Bank of India	Current	Swiggy	Pharmacy	3196.37
3	TXN1000003	CUST70589	Union Bank of India	Salary	BookMyShow	Restaurant	2090.97
4	TXN1000004	CUST28301	Axis Bank	Salary	Jio	Telecom	2747.35
...	...	...	...	...	...	...	...
12802	TXN1012802	CUST83165	Punjab National Bank	Salary	DMart	Restaurant	1504.86
12803	TXN1012803	CUST92904	Bank of Baroda	Business	Airtel	Grocery	7106.29
12804	TXN1012804	CUST43977	ICICI Bank	Salary	Indian Oil	Shopping Mall	4187.23
12805	TXN1012805	CUST37643	Union Bank of India	Salary	Reliance Fresh	Fuel Station	3207.46
12806	TXN1012806	CUST12219	Canara Bank	Savings	Swiggy	Telecom	2868.06

Output columns 2 of 4

	currency	payment_method	transaction_type	...	login_attempts	failed_login_count	previous_transactions
0	INR	IMPS	Recharge	...	4	2	103
1	INR	Credit Card	Purchase	...	1	2	106
2	INR	RTGS	Purchase	...	6	0	77
3	INR	Credit Card	Purchase	...	2	3	50
4	INR	Net Banking	Bill Payment	...	4	3	76
...	...	...	...	...	...	...	...

	currency	payment_method	transaction_type	...	login_attempts	failed_login_count	previous_transactions
12802	INR	IMPS	Recharge	...	2	3	28
12803	INR	Debit Card	Bill Payment	...	3	2	14
12804	INR	Mobile Wallet	Purchase	...	2	3	62
12805	INR	Credit Card	Recharge	...	5	3	100
12806	INR	Mobile Wallet	Ticket Booking	...	2	3	16

Output columns 3 of 4

	average_transaction_amount	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km
0	4313.87	3561.0	10.52	60.11	1770.18
1	2948.79	2575.0	54.85	4.67	2434.39
2	2084.17	2601.0	81.80	9.77	1710.58
3	2518.13	3624.0	58.93	84.23	1124.39
4	2244.80	1918.0	31.69	14.09	2005.49
...	...	...	...	...	...
12802	2028.88	3346.0	34.76	12.99	487.80
12803	9333.58	1088.0	40.24	76.24	1193.40
12804	4700.02	2309.0	74.73	74.36	1765.70
12805	3308.09	2063.0	31.68	29.43	1122.40
12806	3135.45	2333.0	46.84	99.86	2242.15

Output columns 4 of 4

	transaction_duration_sec	fraud_label
0	5.23	Fraudulent
1	156.70	Legitimate
2	56.14	Legitimate
3	31.11	Legitimate
4	91.58	Legitimate
...	...	...
12802	7.01	Legitimate
12803	155.80	Legitimate
12804	54.71	Legitimate
12805	24.09	Fraudulent
12806	183.95	Fraudulent

2.8 Removing Duplicates

```
df.drop_duplicates()
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
0	TXN1000000	CUST46048	Axis Bank	Current	Bharat Petroleum	Pharmacy	3998.30
1	TXN1000001	CUST65392	Canara Bank	Salary	Bharat Petroleum	Restaurant	2706.99
2	TXN1000002	CUST19116	State Bank of India	Current	Swiggy	Pharmacy	3196.37
3	TXN1000003	CUST70589	Union Bank of India	Salary	BookMyShow	Restaurant	2090.97

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
4	TXN1000004	CUST28301	Axis Bank	Salary	Jio	Telecom	2747.35
...	...	...	...	...	...	...	...
12802	TXN1012802	CUST83165	Punjab National Bank	Salary	DMart	Restaurant	1504.86
12803	TXN1012803	CUST92904	Bank of Baroda	Business	Airtel	Grocery	7106.29
12804	TXN1012804	CUST43977	ICICI Bank	Salary	Indian Oil	Shopping Mall	4187.23
12805	TXN1012805	CUST37643	Union Bank of India	Salary	Reliance Fresh	Fuel Station	3207.46
12806	TXN1012806	CUST12219	Canara Bank	Savings	Swiggy	Telecom	2868.06

Output columns 2 of 4

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
0	INR	IMPS	Recharge	...	2	103	4313.87
1	INR	Credit Card	Purchase	...	2	106	2948.79
2	INR	RTGS	Purchase	...	0	77	2084.17
3	INR	Credit Card	Purchase	...	3	50	2518.13
4	INR	Net Banking	Bill Payment	...	3	76	2244.80
...	...	...	...	...	...	...	...
12802	INR	IMPS	Recharge	...	3	28	2028.88
12803	INR	Debit Card	Bill Payment	...	2	14	9333.58
12804	INR	Mobile Wallet	Purchase	...	3	62	4700.02
12805	INR	Credit Card	Recharge	...	3	100	3308.09
12806	INR	Mobile Wallet	Ticket Booking	...	3	16	3135.45

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
0	125217.62	3561.0	10.52	60.11	1770.18	5.23
1	18938.41	2575.0	54.85	4.67	2434.39	156.70
2	772522.81	2601.0	81.80	9.77	1710.58	56.14
3	166749.36	3624.0	58.93	84.23	1124.39	31.11
4	285724.28	1918.0	31.69	14.09	2005.49	91.58
...	...	...	...	...	...	...
12802	119589.43	3346.0	34.76	12.99	487.80	7.01
12803	77331.13	1088.0	40.24	76.24	1193.40	155.80
12804	108584.33	2309.0	74.73	74.36	1765.70	54.71
12805	504981.03	2063.0	31.68	29.43	1122.40	24.09
12806	618620.82	2333.0	46.84	99.86	2242.15	183.95

Output columns 4 of 4

	fraud_label
0	Fraudulent
1	Legitimate
2	Legitimate
3	Legitimate
4	Legitimate

	fraud_label
...	...
12802	Legitimate
12803	Legitimate
12804	Legitimate
12805	Fraudulent
12806	Fraudulent

2.8.1 Checking Duplicates After Cleaning

```
df.duplicated().sum()
```

```
np.int64(0)
```

2.9 Creating Derived Volumns

2.9.1 Transaction Amount Usage Level

```
df['normal_transaction_level']=df['transaction_amount'].apply(
lambda x: 'High' if x >= 70000
else 'Medium' if x>= 50000 else 'Low' )
df
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
0	TXN1000000	CUST46048	Axis Bank	Current	Bharat Petroleum	Pharmacy	3998.30
1	TXN1000001	CUST65392	Canara Bank	Salary	Bharat Petroleum	Restaurant	2706.99
2	TXN1000002	CUST19116	State Bank of India	Current	Swiggy	Pharmacy	3196.37
3	TXN1000003	CUST70589	Union Bank of India	Salary	BookMyShow	Restaurant	2090.97
4	TXN1000004	CUST28301	Axis Bank	Salary	Jio	Telecom	2747.35
...	...	...	...	...	...	...	...
12802	TXN1012802	CUST83165	Punjab National Bank	Salary	DMart	Restaurant	1504.86
12803	TXN1012803	CUST92904	Bank of Baroda	Business	Airtel	Grocery	7106.29
12804	TXN1012804	CUST43977	ICICI Bank	Salary	Indian Oil	Shopping Mall	4187.23
12805	TXN1012805	CUST37643	Union Bank of India	Salary	Reliance Fresh	Fuel Station	3207.46
12806	TXN1012806	CUST12219	Canara Bank	Savings	Swiggy	Telecom	2868.06

Output columns 2 of 4

	currency	payment_method	transaction_type	...	previous_transactions	average_transaction_amount	account_balance
0	INR	IMPS	Recharge	...	103	4313.87	125217.62
1	INR	Credit Card	Purchase	...	106	2948.79	18938.41
2	INR	RTGS	Purchase	...	77	2084.17	772522.81
3	INR	Credit Card	Purchase	...	50	2518.13	166749.36
4	INR	Net Banking	Bill Payment	...	76	2244.80	285724.28
...	...	...	...	...	...	...	...
12802	INR	IMPS	Recharge	...	28	2028.88	119589.43
12803	INR	Debit Card	Bill Payment	...	14	9333.58	77331.13
12804	INR	Mobile Wallet	Purchase	...	62	4700.02	108584.33
12805	INR	Credit Card	Recharge	...	100	3308.09	504981.03
12806	INR	Mobile Wallet	Ticket Booking	...	16	3135.45	618620.82

Output columns 3 of 4

	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec	fraud_label
0	3561.0	10.52	60.11	1770.18	5.23	Fraudulent
1	2575.0	54.85	4.67	2434.39	156.70	Legitimate
2	2601.0	81.80	9.77	1710.58	56.14	Legitimate
3	3624.0	58.93	84.23	1124.39	31.11	Legitimate
4	1918.0	31.69	14.09	2005.49	91.58	Legitimate
...	...	...	...	...	...	...
12802	3346.0	34.76	12.99	487.80	7.01	Legitimate
12803	1088.0	40.24	76.24	1193.40	155.80	Legitimate
12804	2309.0	74.73	74.36	1765.70	54.71	Legitimate
12805	2063.0	31.68	29.43	1122.40	24.09	Fraudulent
12806	2333.0	46.84	99.86	2242.15	183.95	Fraudulent

Output columns 4 of 4

	normal_transaction_level
0	Low
1	Low
2	Low
3	Low
4	Low
...	...
12802	Low
12803	Low
12804	Low
12805	Low
12806	Low

2.10 Filtering Data

2.10.1 Filtering Average_Transaction_Amount Above 4000

```
df[df['average_transaction_amount'] > 4000][['average_transaction_amount']]
```

	average_transaction_amount
0	4313.87
7	4110.86
13	5414.26
15	5720.81
17	16079.54
...	...
12798	4191.49
12799	6527.07
12801	4444.12
12803	9333.58

	average_transaction_amount
12804	4700.02

2.10.2 Filtration Based on Merchant_Risk_Score and Ip_Risk_Score

```
filtered_data = df[
    (df['merchant_risk_score'] > 3000) &
    (df['ip_risk_score'] > 50000)
]
filtered_data
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount	currency
--	----------------	-------------	-----------	--------------	---------------	-------------------	--------------------	----------

Output columns 2 of 4

	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
--	----------------	------------------	-----	--------------------	-----------------------	----------------------------

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
--	-----------------	----------------------	---------------------	---------------	-------------------------	--------------------------

Output columns 4 of 4

	fraud_label
--	-------------

2.11 Aggregating Data

2.11.1 Count on Failed_Login_Count

```
df['failed_login_count']
```

	failed_login_count
0	2
1	2
2	0
3	3
4	3
...	...
12802	3
12803	2
12804	3
12805	3
12806	3

2.11.2 Count of Fraudulent Average Transaction Amount

```
df['average_transaction_amount'].value_counts()
```

	count
average_transaction_amount	
2694.25	3
710.01	2

	count
2346.99	2
3177.75	2
1426.88	2
...	...
4127.60	1
1912.95	1
2791.33	1
314.07	1
5228.22	1

2.11.3 Finding the Average Transaction Amount

```
df['average_transaction_amount'].mean()
```

```
np.float64(3352.8384032169906)
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
10418	TXN1010418	CUST83145	HDFC Bank	Business	BSNL	Restaurant	4023.30
6176	TXN1006176	CUST28315	Canara Bank	Savings	Tata Power	Utility Bill	2493.47
11056	TXN1011056	CUST66252	Union Bank of India	Salary	Tata Power	Pharmacy	5087.15
6878	TXN1006878	CUST64121	State Bank of India	Salary	Airtel	Travel	2813.53
5788	TXN1005788	CUST74201	Canara Bank	Current	Reliance Fresh	Shopping Mall	6478.20
...	...	...	...	...	...	...	...
9967	TXN1009967	CUST54910	Axis Bank	Salary	DMart	Grocery	4442.84
9513	TXN1009513	CUST65322	Punjab National Bank	Business	IRCTC	Education	3404.98
919	TXN1000919	CUST33356	State Bank of India	Current	Airtel	Education	488.16
4913	TXN1004913	CUST54104	Indian Bank	Business	Zomato	Shopping Mall	4544.02
11407	TXN1011407	CUST59846	HDFC Bank	Business	Zomato	Education	605.55

Output columns 2 of 4

	currency	payment_method	transaction_type	...	failed_login_count	previous_transactions	average_transaction_amount
10418	INR	IMPS	Money Transfer	...	3	50	2726.68
6176	INR	Net Banking	Bill Payment	...	1	10	2223.09
11056	INR	IMPS	Recharge	...	0	9	4981.86
6878	INR	IMPS	Recharge	...	2	99	2975.07
5788	INR	UPI	Purchase	...	0	95	8828.03
...	...	...	...	...	...	...	...
9967	INR	Credit Card	Money Transfer	...	3	46	6151.96
9513	INR	Mobile Wallet	Bill Payment	...	2	82	2530.18
919	INR	Credit Card	Recharge	...	1	43	584.30
4913	INR	Credit Card	Recharge	...	2	66	3020.58
11407	INR	UPI	Purchase	...	3	101	550.57

Output columns 3 of 4

	account_balance	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec
10418	505.25	1061.0	38.37	72.28	913.85	128.05
6176	580.68	1978.0	35.16	2.86	506.86	182.65
11056	638.10	636.0	61.71	67.04	2234.82	144.43
6878	642.72	3518.0	14.91	10.94	516.91	17.54
5788	644.15	498.0	50.30	66.50	1758.22	222.19
...	...	...	...	...	...	...
9967	799810.24	1788.0	22.37	61.16	1857.69	174.57
9513	799852.44	821.0	61.01	22.24	1764.84	125.55
919	799853.47	1336.0	55.68	99.02	936.68	254.31
4913	799914.39	2833.0	26.24	13.26	2204.42	83.99
11407	799962.12	2242.0	45.79	81.07	2042.76	272.62

Output columns 4 of 4

	fraud_label
10418	Legitimate
6176	Legitimate
11056	Legitimate
6878	Legitimate
5788	Legitimate
...	...
9967	Legitimate
9513	Legitimate
919	Legitimate
4913	Legitimate
11407	Legitimate

3. Exploratory Data Analysis (EDA)

3.1 Finding Maximum of Merchant Risk Score

```
df['merchant_risk_score'].max()
```

100.0

3.2 Finding Minimum of merchant_risk_score

```
df['merchant_risk_score'].min()
```

5.0

3.3 Sorting Account Balance in Ascending Order

```
df.sort_values('account_balance')
```

Output columns 1 of 4

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
10418	TXN1010418	CUST83145	HDFC Bank	Business	BSNL	Restaurant	4023.30
6176	TXN1006176	CUST28315	Canara Bank	Savings	Tata Power	Utility Bill	2493.47
11056	TXN1011056	CUST66252	Union Bank of India	Salary	Tata Power	Pharmacy	5087.15
6878	TXN1006878	CUST64121	State Bank of India	Salary	Airtel	Travel	2813.53

	transaction_id	customer_id	bank_name	account_type	merchant_name	merchant_category	transaction_amount
5788	TXN1005788	CUST74201	Canara Bank	Current	Reliance Fresh	Shopping Mall	6478.20
...	...	...	...	...	...	...	...
9967	TXN1009967	CUST54910	Axis Bank	Salary	DMart	Grocery	4442.84
9513	TXN1009513	CUST65322	Punjab National Bank	Business	IRCTC	Education	3404.98
919	TXN1000919	CUST33356	State Bank of India	Current	Airtel	Education	488.16
4913	TXN1004913	CUST54104	Indian Bank	Business	Zomato	Shopping Mall	4544.02
11407	TXN1011407	CUST59846	HDFC Bank	Business	Zomato	Education	605.55

Output columns 2 of 4

	currency	payment_method	transaction_type	...	previous_transactions	average_transaction_amount	account_balance
10418	INR	IMPS	Money Transfer	...	50	2726.68	505.25
6176	INR	Net Banking	Bill Payment	...	10	2223.09	580.68
11056	INR	IMPS	Recharge	...	9	4981.86	638.10
6878	INR	IMPS	Recharge	...	99	2975.07	642.72
5788	INR	UPI	Purchase	...	95	8828.03	644.15
...	...	...	...	...	...	...	...
9967	INR	Credit Card	Money Transfer	...	46	6151.96	799810.24
9513	INR	Mobile Wallet	Bill Payment	...	82	2530.18	799852.44
919	INR	Credit Card	Recharge	...	43	584.30	799853.47
4913	INR	Credit Card	Recharge	...	66	3020.58	799914.39
11407	INR	UPI	Purchase	...	101	550.57	799962.12

Output columns 3 of 4

	beneficiary_age_days	merchant_risk_score	ip_risk_score	transaction_distance_km	transaction_duration_sec	fraud_label
10418	1061.0	38.37	72.28	913.85	128.05	Legitimate
6176	1978.0	35.16	2.86	506.86	182.65	Legitimate
11056	636.0	61.71	67.04	2234.82	144.43	Legitimate
6878	3518.0	14.91	10.94	516.91	17.54	Legitimate
5788	498.0	50.30	66.50	1758.22	222.19	Legitimate
...	...	...	...	...	...	...
9967	1788.0	22.37	61.16	1857.69	174.57	Legitimate
9513	821.0	61.01	22.24	1764.84	125.55	Legitimate
919	1336.0	55.68	99.02	936.68	254.31	Legitimate
4913	2833.0	26.24	13.26	2204.42	83.99	Legitimate
11407	2242.0	45.79	81.07	2042.76	272.62	Legitimate

Output columns 4 of 4

	normal_transaction_level
10418	Low
6176	Low
11056	Low
6878	Low
5788	Low

	normal_transaction_level
...	...
9967	Low
9513	Low
919	Low
4913	Low
11407	Low

3.4 Finding Mean of Transaction Amount Grouped By Merchant Category Using Group By

```
df.groupby('merchant_category')['transaction_amount'].mean()
```

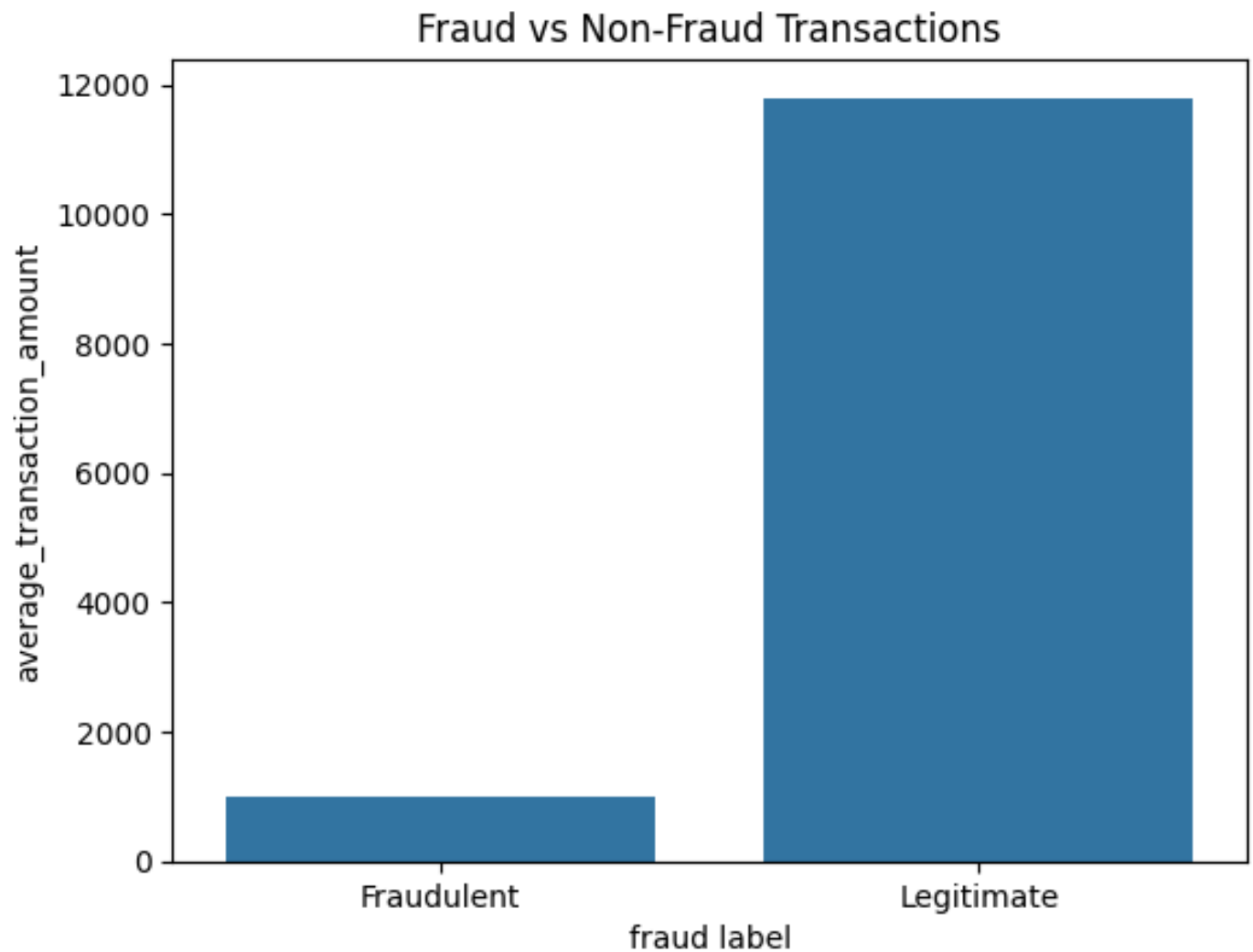
	transaction_amount
merchant_category	
E-Commerce	3417.385489
Education	3282.010042
Fuel Station	3351.113093
Grocery	3433.358349
Hospital	3425.044666
Pharmacy	3253.159038
Restaurant	3441.149830
Shopping Mall	3324.326942
Telecom	3369.097043
Travel	3236.416368
Utility Bill	3463.271380

4.0 Visualisation

```
import matplotlib.pyplot as plt
import seaborn as sns
```

4.1 Bar Chart Showing the Number of Fraudulent and Non-Fraudulent Transactions

```
sns.countplot(x='fraud_label',data=df)
plt.title('Fraud vs Non-Fraud Transactions')
plt.xlabel('fraud_label')
plt.ylabel('average_transaction_amount')
plt.show()
```

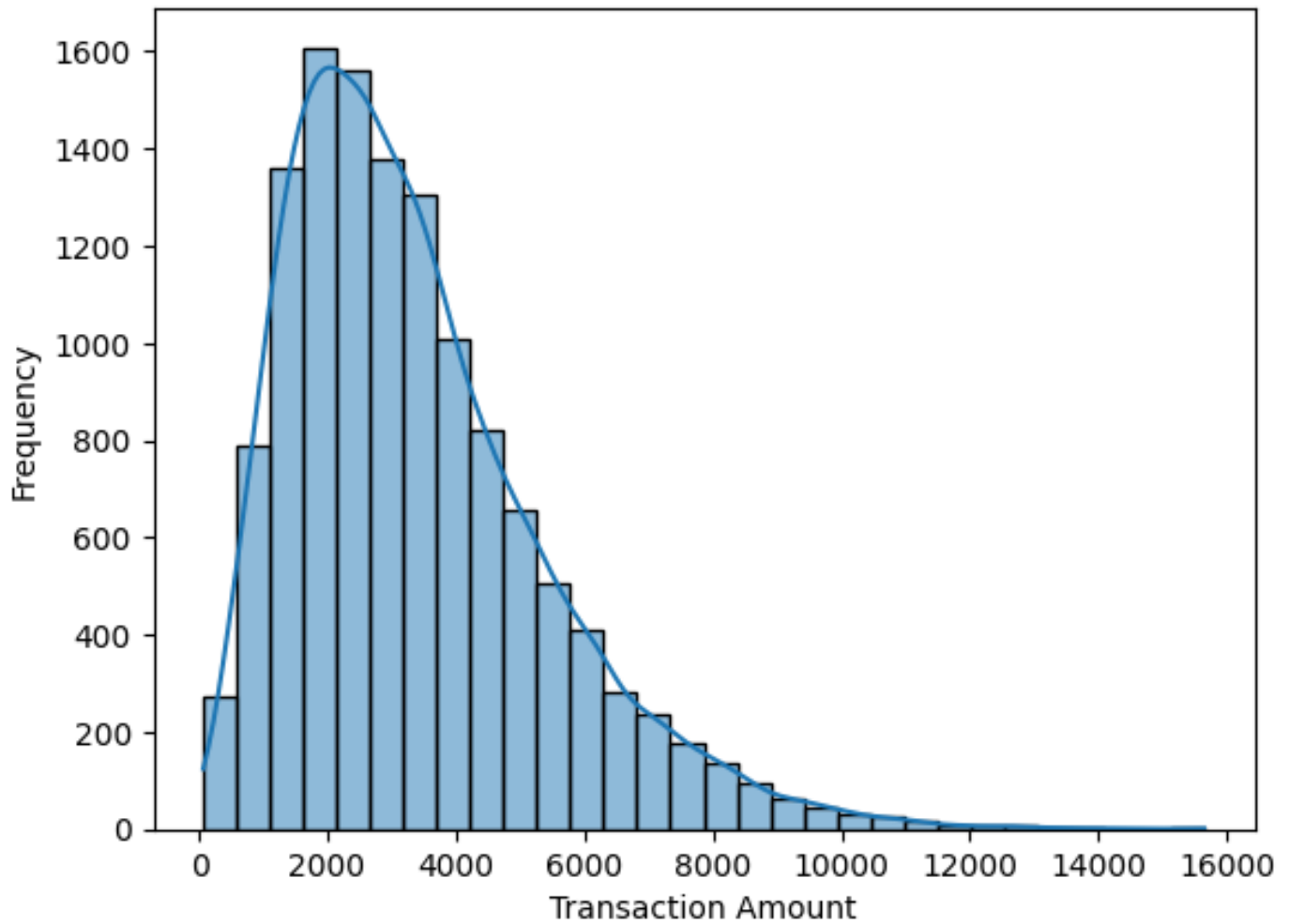


4.2 Distribution of Transaction Amount

Histogram Showing How Transaction Amounts are Distributed and Whether Most Transactions are Low or High Value.

```
sns.histplot(data=df, x='transaction_amount', bins=30, kde=True)
plt.title('Distribution of Transaction Amount')
plt.xlabel('Transaction Amount')
plt.ylabel('Frequency')
plt.show()
```

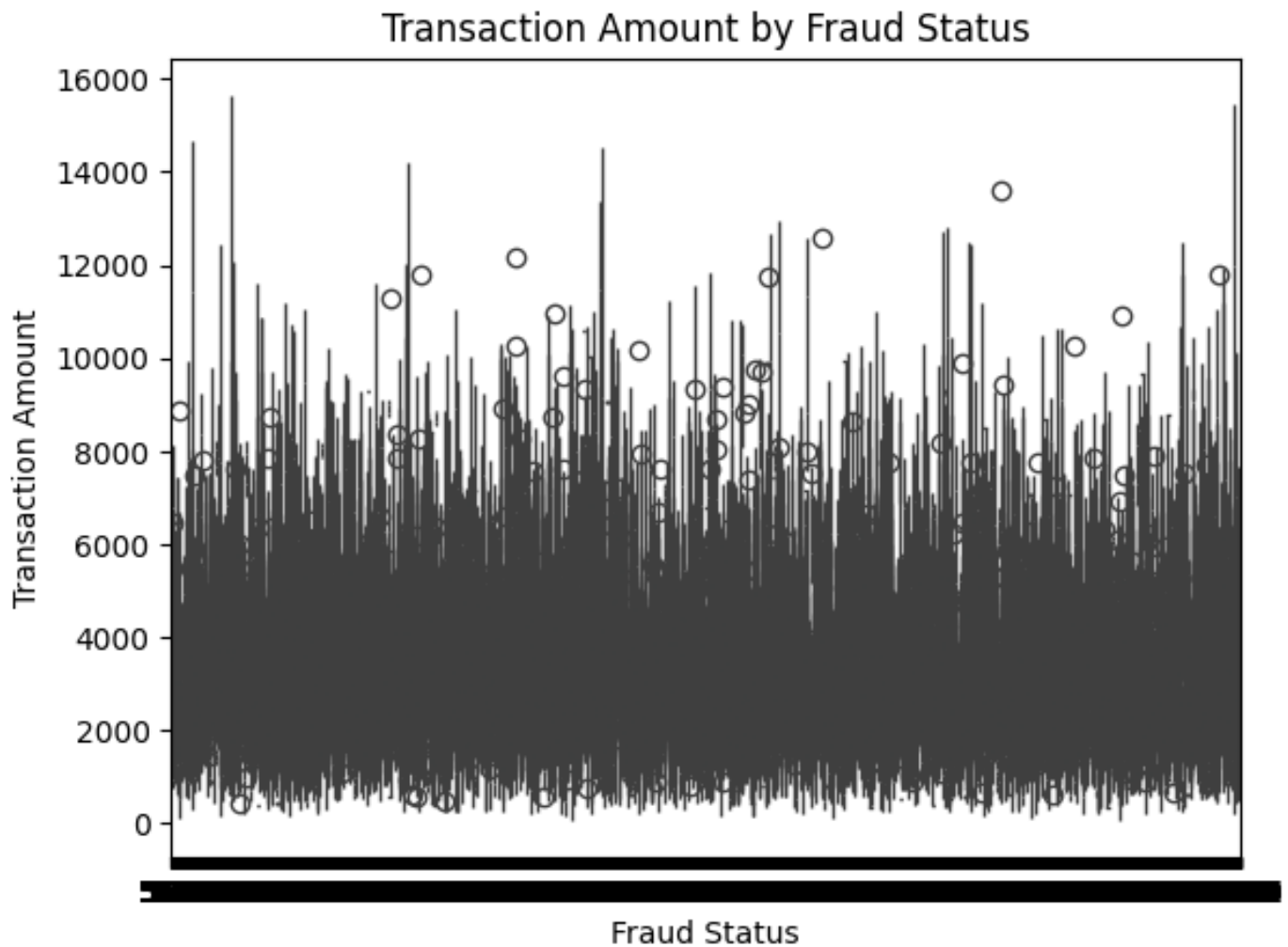
Distribution of Transaction Amount



4.3 BOXPLOT

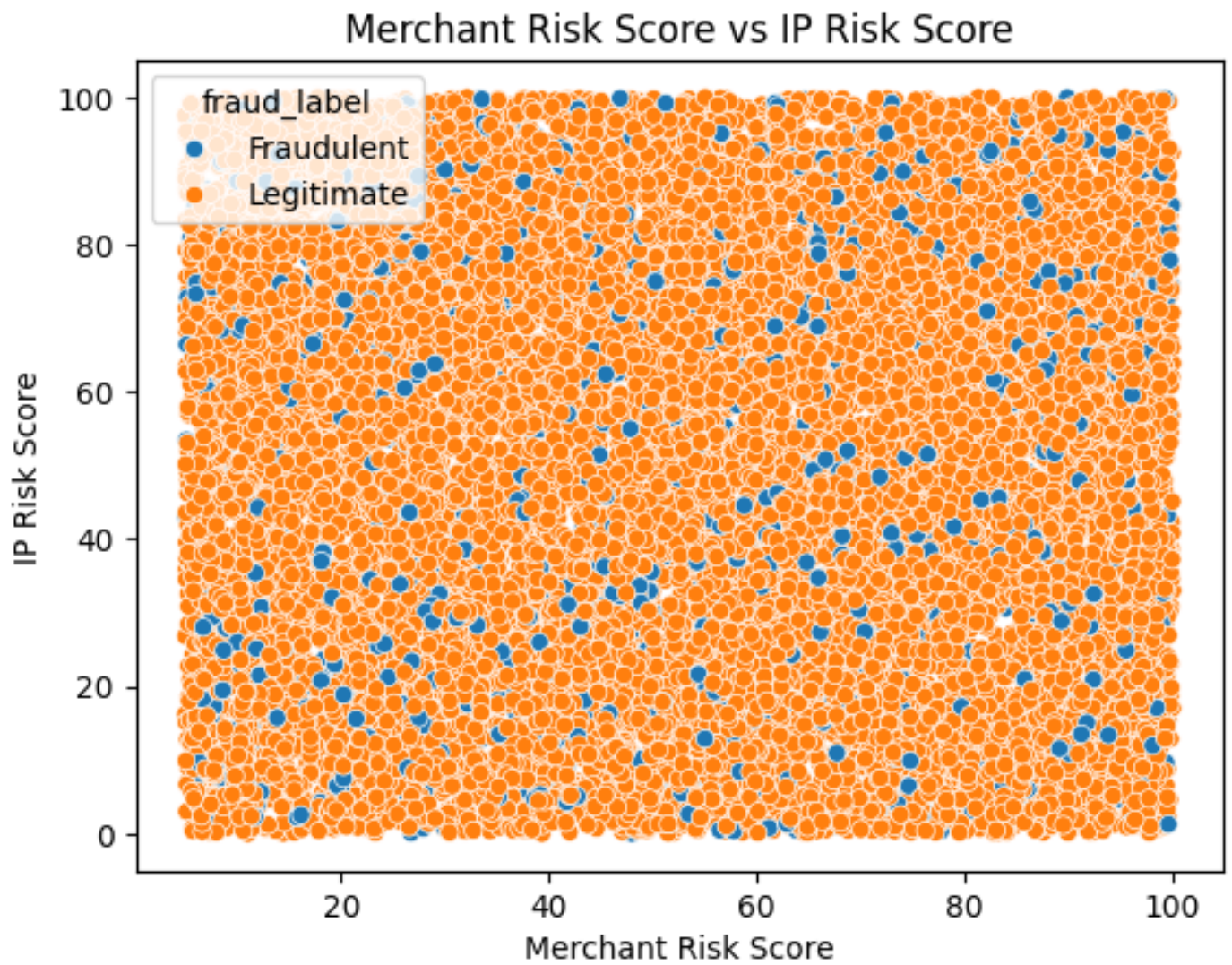
Boxplot Comparing Transaction Amounts for Fraudulent and Non-Fraudulent Transactions.

```
sns.boxplot(data=df, x='merchant_risk_score', y='transaction_amount')
plt.title('Transaction Amount by Fraud Status')
plt.xlabel('Fraud Status')
plt.ylabel('Transaction Amount')
plt.show()
```



4.4 Scatterplot Showing Merchant Risk Score vs IP Risk Score

```
sns.scatterplot(data=df, x='merchant_risk_score', y='ip_risk_score', hue='fraud_label')  
plt.title('Merchant Risk Score vs IP Risk Score')  
plt.xlabel('Merchant Risk Score')  
plt.ylabel('IP Risk Score')  
plt.show()
```

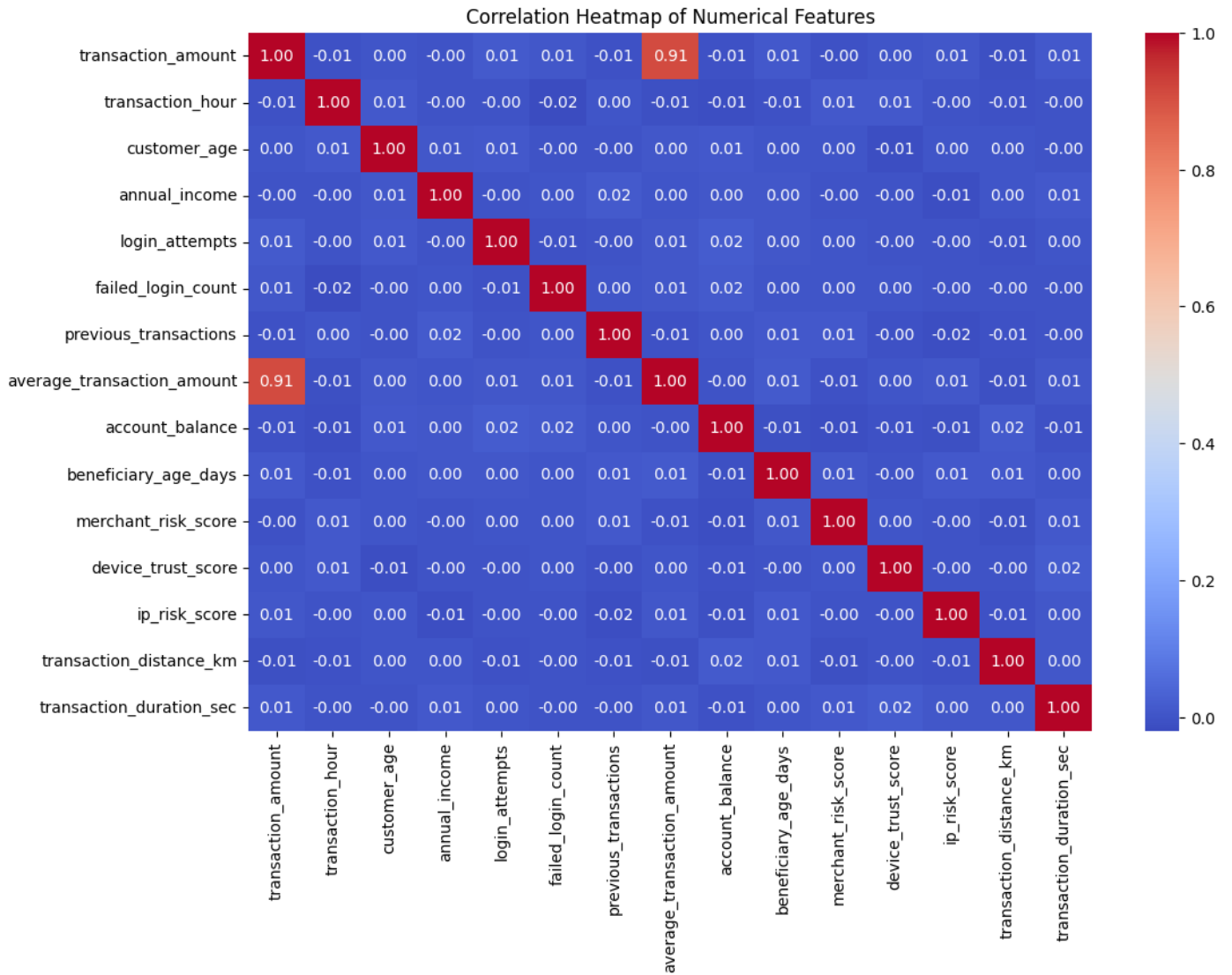


4.5 Correlation Heatmap

```
plt.figure(figsize=(12, 8))

sns.heatmap(
    df.select_dtypes(include='number').corr(),
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

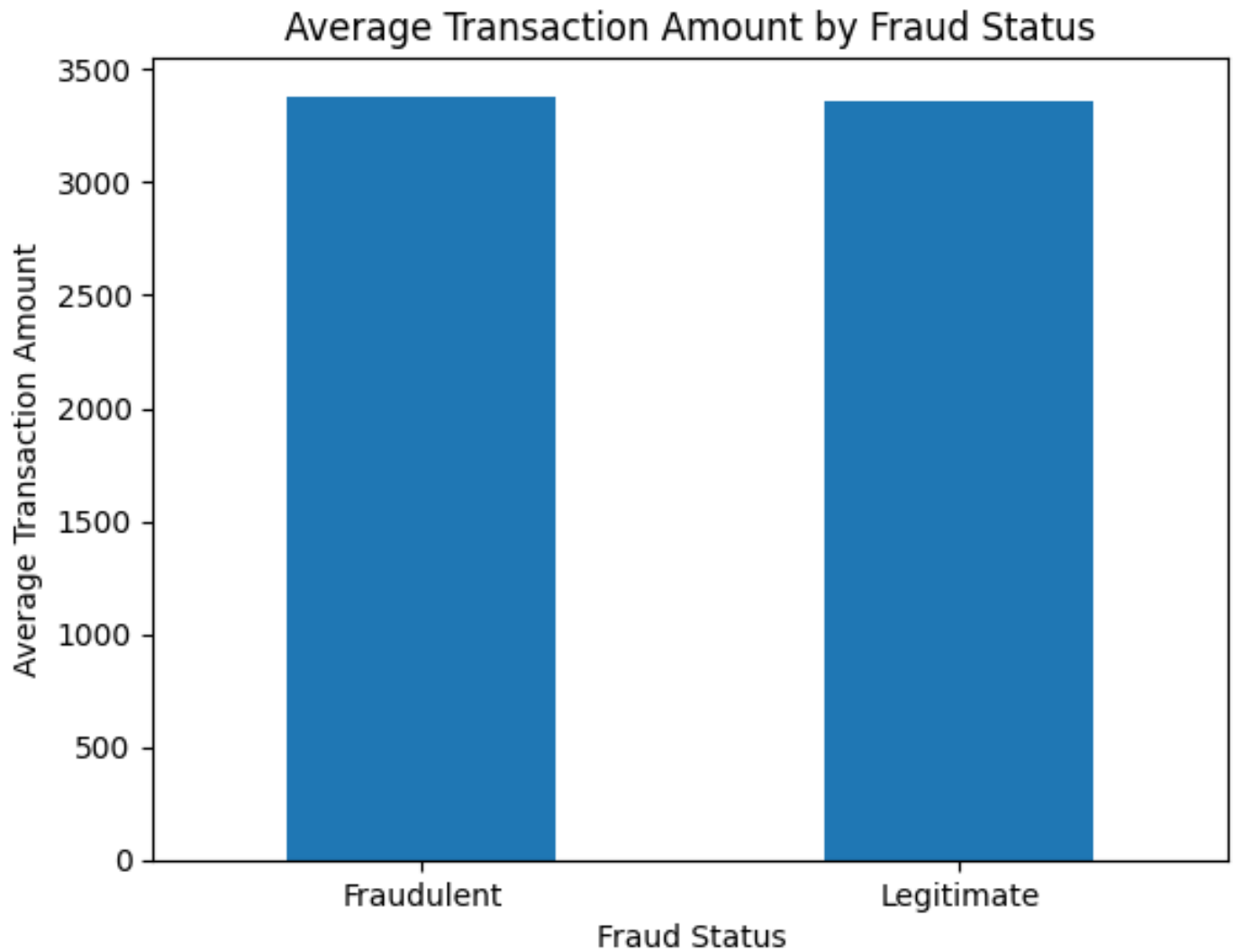
plt.title('Correlation Heatmap of Numerical Features')
plt.show()
```



4.6 Bar Graph Showing Average Transaction Amount by Fraud Status

```
avg_amount = df.groupby('fraud_label')['transaction_amount'].mean()

avg_amount.plot(kind='bar')
plt.title('Average Transaction Amount by Fraud Status')
plt.xlabel('Fraud Status')
plt.ylabel('Average Transaction Amount')
plt.xticks(rotation=0)
plt.show()
```

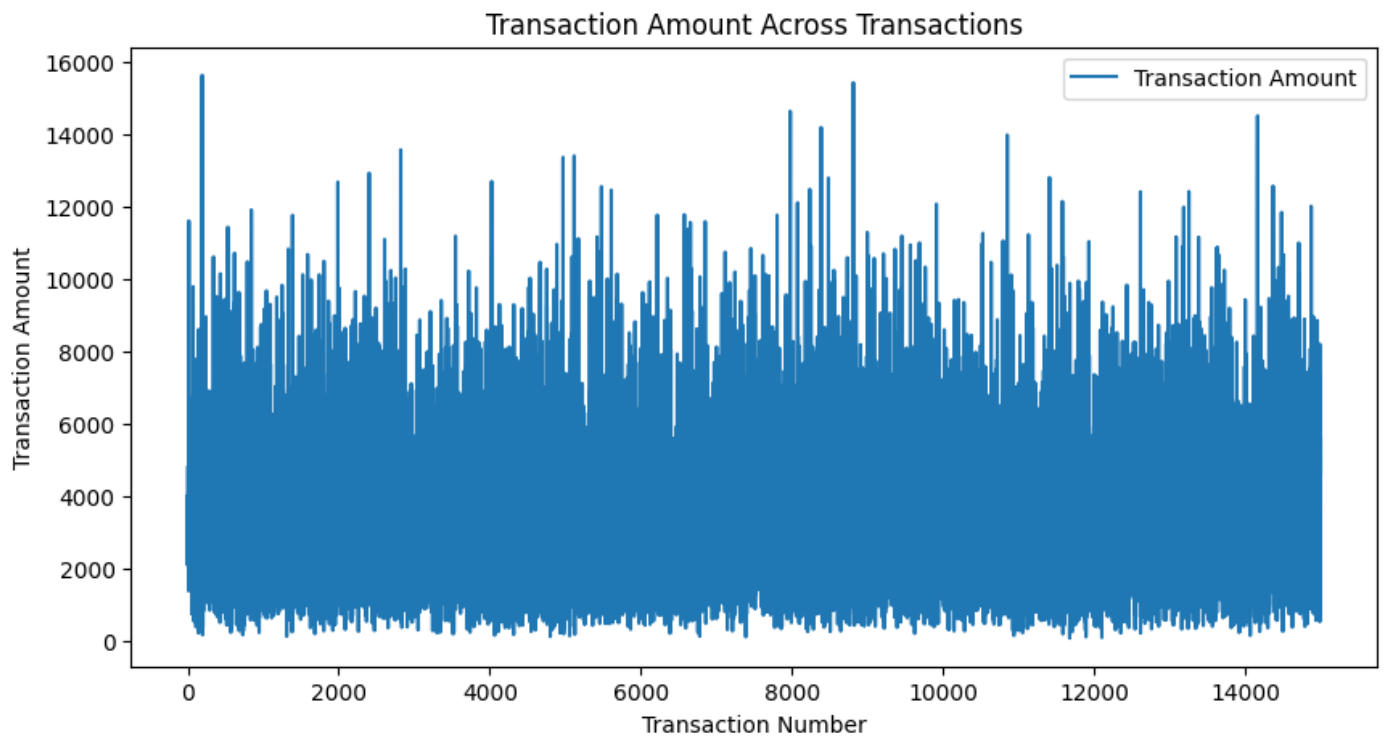


4.7 Line plot to show how the average transaction amount changes across transaction records.

```
plt.figure(figsize=(10, 5))

plt.plot(
    df.index,
    df['transaction_amount'],
    label='Transaction Amount'
)

plt.title('Transaction Amount Across Transactions')
plt.xlabel('Transaction Number')
plt.ylabel('Transaction Amount')
plt.legend()
plt.show()
```



4.8 Pie Chart Showing the Percentage of Fraudulent VS Non-Fraudulent Transactions.

```

fraud_counts = df['fraud_label'].value_counts()

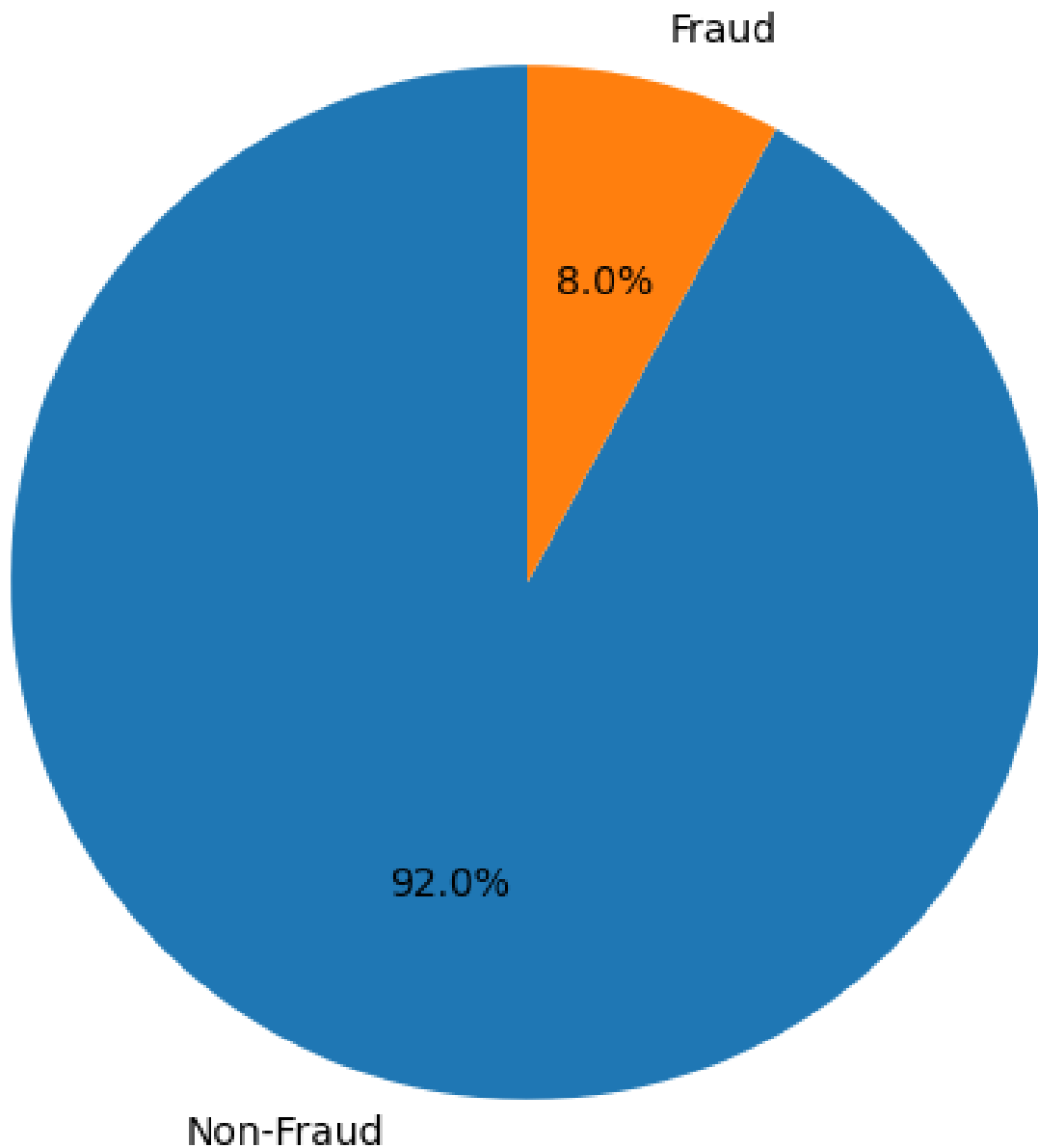
plt.figure(figsize=(6, 6))

plt.pie(
    fraud_counts,
    labels=['Non-Fraud', 'Fraud'],
    autopct='%1.1f%%',
    startangle=90
)

plt.title('Fraud vs Non-Fraud Transactions')
plt.show()

```

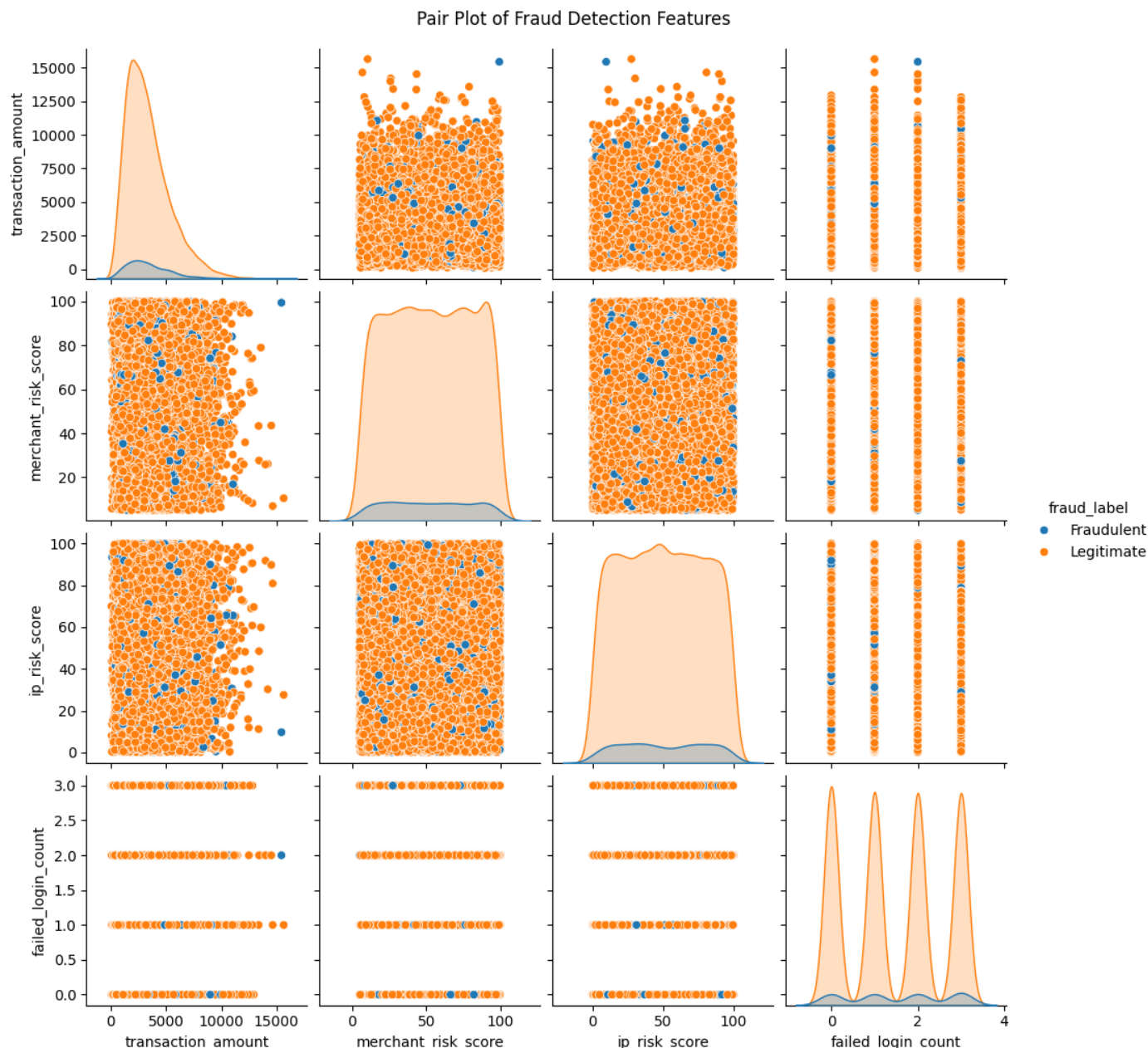
Fraud vs Non-Fraud Transactions



4.9 Pair Plot to Show the Relationships Between Several Numerical Variables and How They Differ Between Fraud and Non-Fraud Transactions.

```
sns.pairplot(
    df[['transaction_amount',
        'merchant_risk_score',
        'ip_risk_score',
        'failed_login_count',
        'fraud_label']],
    hue='fraud_label'
)

plt.suptitle('Pair Plot of Fraud Detection Features', y=1.02)
plt.show()
```



5 Project Summary

The **Fraud Detection Using Python** project focuses on analyzing financial transaction data to identify patterns and characteristics associated with fraudulent transactions. The project uses **Python, Pandas, Matplotlib, and Seaborn** for data loading, cleaning, exploration, analysis, and visualization.

The dataset contains transaction-related features such as transaction amount, merchant risk score, IP risk score, failed login count, and fraud status. Exploratory Data Analysis (EDA) was performed to understand the dataset, identify patterns, compare fraudulent and non-fraudulent transactions, and examine relationships between important variables.

Various visualizations, including **bar charts, histograms, box plots, scatter plots, pair plots, line plots, pie charts, and correlation heatmaps**, were used to present the findings clearly. The analysis helps identify factors that may be associated with fraudulent transactions and provides useful insights for further fraud detection and predictive modeling.

6 Project Overview

The objective of this project is to analyze a **Fraud Detection Dataset** using Python and understand the factors that can help distinguish fraudulent transactions from legitimate transactions.

The project follows a structured data analysis process:

1. **Data Loading** - The dataset was imported into Python using Pandas.
2. **Data Understanding** - The shape, columns, data types, and basic statistical information of the dataset were examined.
3. **Data Cleaning** - Missing values, inconsistent data, and other data-quality issues were identified and handled where required.
4. **Data Filtering and Analysis** - Transactions were filtered based on different conditions, such as transaction amount, merchant risk score, IP risk score, and failed login count.
5. **Exploratory Data Analysis** - Numerical and categorical variables were analyzed to identify patterns and relationships.
6. **Data Visualization** - Matplotlib and Seaborn were used to create different charts, including count plots, histograms, box plots, scatter plots, pair plots, line plots, pie charts, and heatmaps.
7. **Fraud Pattern Analysis** - Fraudulent and non-fraudulent transactions were compared to understand differences in transaction characteristics and risk-related factors.

This project demonstrates how Python-based data analysis and visualization can be used to transform raw transaction data into meaningful insights that can support fraud investigation and future machine-learning applications.

7 Conclusion

The Fraud Detection project demonstrates the importance of **data analysis and visualization in identifying potential fraud patterns** in financial transactions. Through data cleaning, filtering, statistical analysis, and visualization, important transaction characteristics such as transaction amount, merchant risk score, IP risk score, and failed login count were explored.

The visualizations provided a clear understanding of the differences between fraudulent and non-fraudulent transactions and helped identify relationships and unusual patterns within the dataset.

Overall, the project provided practical experience in using **Pandas for data manipulation and analysis, Matplotlib and Seaborn for visualization**, and demonstrated how exploratory data analysis can serve as an important foundation for developing more advanced **fraud detection and machine-learning models** in the future.